

# The camera, worked through

What a 6 mm lens on an IMX477 actually sees, and why the sizing chapter disagrees

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## 1. Why I went back to this

We are buying an Arducam IMX477 with a 6 mm S-mount lens, and I wanted the line spacing and sweep times to come out of the sensor rather than out of a table someone had already written. Working it through, the two do not agree, and the reason is worth writing down.

Sony specifies the IMX477 as a type 1/2.3 sensor:  $4056 \times 3040$  pixels on a **1.55  $\mu\text{m}$**  pitch, 7.9 mm across the diagonal. Our own sizing-calculations.md §8 gives the baseline as “1/1.8 in,  $4056 \times 3040$ ,  $7.4 \times 5.6$  mm, **1.82  $\mu\text{m}$**  pitch”.

**The pixel count is identical and the pixel size is not.** Every optical quantity we depend on scales with pitch, so the field of view, ground sample distance, swath, line spacing and sweep time in that chapter all describe a camera we are not buying. What follows derives both, so the difference can be seen rather than argued about.

## 2. From three numbers to a field of view

Everything here comes from the pixel count, the pixel pitch and the focal length. The active area is simply count times pitch,

$$w = N_x p = 4056 \times 1.55 \mu\text{m} = 6.287 \text{ mm}, \quad h = N_y p = 4.712 \text{ mm},$$

giving a 7.86 mm diagonal, which is the type 1/2.3 format Sony quotes. The lens forms that image at its focal length, so each half-field is the arctangent of half the sensor over  $f$ . There is no small-angle approximation in this and there does not need to be:

$$\theta = 2 \arctan\left(\frac{s}{2f}\right) \implies \text{HFOV} = 55.3^\circ, \quad \text{VFOV} = 42.9^\circ, \quad \text{DFOV} = 66.4^\circ.$$

A single pixel subtends 14.80 mdeg. That number returns in §5, because it is what decides how long the shutter may stay open.

## 3. Ground sample distance, footprint and spacing

The sensor and the ground form similar triangles about the lens, which gives the ground sample distance directly, and the swath follows from the field of view. Line spacing is the swath less the sidelap:

$$\text{GSD} = \frac{pH}{f}, \quad W = 2H \tan \frac{\text{HFOV}}{2}, \quad S = W(1 - \ell), \quad \ell = 30\%.$$

Sidelap is not conservatism for its own sake. Attitude wobble, altitude error and lens distortion all move the real footprint around between one pass and the next, and a gap between passes is not a cosmetic defect — it is a survivor nobody flew over.

| AGL       | GSD         | Survivor   | Footprint                            | Spacing     | Lines    | Sweep      |
|-----------|-------------|------------|--------------------------------------|-------------|----------|------------|
| (m)       | (cm/px)     | (px)       | (m)                                  | (m)         |          | (s)        |
| 30        | 0.78        | 219        | $31.4 \times 23.6$                   | 22.0        | 9        | 253        |
| <b>40</b> | <b>1.03</b> | <b>165</b> | <b><math>41.9 \times 31.4</math></b> | <b>29.3</b> | <b>7</b> | <b>196</b> |
| 60        | 1.55        | 110        | $62.9 \times 47.1$                   | 44.0        | 5        | 138        |

Transect counts and sweep times take the 10 ha area divided 3 ways, treat each sub-region as square, and fly it at 8 m/s with six seconds spent in each turn.

#### 4. What changed when the sensor changed

Comparing the two at the 40 m search altitude shows the substitution is not simply better or worse. It is a trade, and it happens to be one nobody made deliberately.

| At 40 m AGL                      | IMX477 | Sizing chapter | Change, % |
|----------------------------------|--------|----------------|-----------|
| Horizontal field of view, °      | 55.30  | 63.20          | -12.5     |
| Ground sample distance, cm/px    | 1.03   | 1.21           | -14.8     |
| Survivor along the long axis, px | 164.52 | 140.11         | +17.4     |
| Swath width, m                   | 41.91  | 49.21          | -14.8     |
| Line spacing, m                  | 29.34  | 34.45          | -14.8     |
| Transects per drone              | 7.00   | 6.00           | +16.7     |
| Sweep time per drone, s          | 195.75 | 166.93         | +17.3     |

A smaller pixel behind the same lens narrows the field and finens the ground sample distance in exactly the same proportion, so the two effects are the same number wearing different signs. **We gain 17 % more pixels on a survivor and pay 17 % more sweep time** — seven transects per drone instead of six.

Set against 250 marks for detection and an 1800 s mission budget, I think that trade favours the part we are actually buying. The uncomfortable part is that it was never chosen: the figures we have been quoting describe neither camera.

#### 5. Two things this settles

**Motion blur is not the constraint we half-expected.** Forward motion smears the image by  $v t_{\text{exp}}$  on the ground, so holding that under a single pixel means  $t_{\text{exp}} \leq \text{GSD}/v$ :

| AGL (m) | Longest exposure (ms) | Shutter |
|---------|-----------------------|---------|
| 30      | 0.97                  | 1/1032  |
| 40      | 1.29                  | 1/774   |
| 60      | 1.94                  | 1/516   |

SYS-45 already requires 1/1000 s or faster, which clears every altitude with room to spare. In Indian daylight that requirement drives ISO down rather than exposure up, so it costs us nothing.

**The tiling budget survives, but one published figure does not.** Tile count follows pixel count, which did not change, so the inference load is untouched:

| Downsample | Frame       | Tiles | Inferences/s | GSD (cm/px) | Survivor (px) |
|------------|-------------|-------|--------------|-------------|---------------|
| 1×         | 4056 × 3040 | 48    | 240          | 1.03        | 165           |
| 2×         | 2028 × 1520 | 12    | 24           | 2.07        | 82            |

At 2× downsampling and 2 Hz we still get 24 inferences per second. But the survivor in that downsampled frame is **82 px**, not the 47 px the proposal currently states — that figure was computed at 60 m, on the other sensor, and both of those need correcting together.

## 6. What I would do about it

Nothing here is an argument for changing the camera. The IMX477 is the better of the two where the marks are, and it is already on order.

It is an argument for re-baselining the documents onto the part we are buying: the GSD table, the swath and line-spacing figures, the sweep times, and the survivor pixel count in the perception section. That work runs into the unresolved question of whether we survey at 40 m or 60 m, because both move the same quantities, and I would rather settle the two in one pass than do the arithmetic twice.

The prose here is mine; the arithmetic is not. Every figure in this note is computed by `camera_optics.py` and typeset by `camera_optics_tex.py`, both under `tools/sizing-model/`, so the document cannot drift from the model the way the sizing chapter drifted from the parts list — which is the whole subject of the note.